Max COPPIN

PhD Student

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Work Experience

Feb and Jun 2025 — **Academic Visitor, AOPP, Oxford University** , *Oxford*, England
- Supervisors: Peter READ and Bruno DEREMBLE
- Project: Turbulent heat flux observation in convection experiment

Now Oct 2023 — **PhD, Institut des Géosciences de l'Environnement (IGE)**, *Grenoble*, France
- Supervisors: Joel SOMMERIA and Bruno DEREMBLE
- Project: Observation of turbulence and the dynamics of the ocean mixed layer in rotation

Apr 2023 Sept 2023 — **Research internship, Laboratoire d'Océanographie Physique et Spatiale (LOPS)**, *Plouzané*, France
- Supervisors: Xavier CARTON and Camille LIQUE
- Project: Stability of mesoscale vortex under sea ice friction in the Arctic

Apr 2023 Sept 2023 — **Research internship, LOPS**, *Plouzané,*, France
- Supervisors: Xavier CARTON and Pierre L'HÉGARET
- Project: Front dynamics, between the inflowing Indian Ocean Surface Water, and the outflowing Persian Gulf water, from the observations of 2014

Education

Now–2023 — **PhD in Physical Oceanography**, *LEGI,IGE*, Grenoble

2023–2021 — **Master in Marine Physics**, *University of Western Brittany*, Brest,
M.Sc with highest Honors, rank 2
Specialisation in Ocean and Climate Physics

2021–2020 — **3rd year undergraduate student in Fundamental Mathematics**,
University of Western Brittany, Brest, *B.Sc with Honors, rank 3*
Option in Arithmetic and combinatory applications and graphs

Publication in peer-reviewed journals

1. **Coppin, M.**, Deremble, B., Sommeria, J. Valran ,T., Viboud, T (submitted.) Laboratory observations of a transient Ekman layer *Geophysical and Astrophysical Fluid Dynamics*

2. Marchand, O., Negretti, E., **Coppin, M.**. Valran ,T., Viboud, T., Deremble, B., Sommeria, J. (In prep.) Laboratory experiments on rotating convection in a stratified fluid at high Rayleigh number *Journal of Fluid Mechanics*

3. **Coppin, M.**, Deremble, B., Sommeria, J. (2025) Wind-Mixed Layer Deepening in a Rotating Frame *Geophysical and Astrophysical Fluid Dynamics*, 139. https://doi.org/10.1080/03091929.2025.2609460

Research Interests

Ekman Dynamics, air-ice-sea interactions, sea-ice formation and melting, ocean dynamics in the polar basin, turbulence in convection, high resolution numerical modelling, experimental geophysical fluid dynamic.

Grants - Funding

- Sept 2024 - International mobility grant, **IDEX, Grenoble, France**, €3053
- Apr 2023 - Thematic Schools funding, **ISBlue, Brest, France**, €1300

Participation in scientific events

Workshop

May 2026 **Theoretical Challenges in Climate Science**, *Paris*, France, Oral presentation

Jan 2026 **DRAKKAR Ocean Modelling Workshop**, *Grenoble*, France, Poster

June 2025 **International Conference on Geophysical and Astrophysical Fluid**, *Plouzané*, France, Oral presentation

Mar 2025 **Parameterizations for global dynamical models in Climatology, Astrophysics and Planetology**, *Les Houches*, France, Oral presentation

May 2024 **Interfaces in the climate system**, *Grenoble*, France, Oral presentation

Nov 2023 **Convection's Days**, *Avignon*, France, Oral presentation

Thematic school

Nov 2025 **Theoretical Challenges for ocean dynamics**, *ENS Lyon*, France

Jan 2024 **Introduction to data Assimilation**, *Université de grenoble alpes*, France

Apr 2023 **Fluid mechanics of planets and stars**, *CISM - Udine*, Italy

Event organisation

Mar 2026 **Early career GDR MFGA workshop**, *ENS Lyon*, France, Co-organising a one-day workshop, (45 participants)

Skills

Numerical skills

- **Programming languages:**
  - Python (advanced): data analysis, post- processing *Jupyter notebooks / VScode*
  - MATLAB (advanced): data analysis, post-processing
  - Fortran (advanced): code setup, modification, and debugging
  - Bash (working knowledge): SSH, job launching, basic workflow management
  - Julia (basic knowledge)
- **Computing environments:** Linux (working knowledge)
- **Ocean modelling:**
  - GOTM: (setup, configuration, runs, output analysis)
  - CROCO (setup, runs, output analysis)
  - DEDALUS (runs)
  - Oceananigans (runs)
- **Version control:** Git
- **Post Processing environment:**
  - UVMAT (Experimental post processing, PIV/LIF, workflows on cluster)

Experimental skills

- **Large-scale rotating facility (Coriolis platform, 13 m):** Coordinated experimental planning and design for forced and free convection experiment
- **Rotating tank experiment (horizontal convection):** Set up and instrumented the apparatus. Implemented a thermistor-based temperature acquisition.
- **Cubic tank (1 m^3- thermal plume):** Built the setup. Tested a new **T-LIF** temperature acquisition technique.

Teaching & Supervision

2026 **University of Grenoble Alpes, Grenoble, France**, *3rd year undergraduate*, Introduction to Physical Oceanography - 12h
Oceanic circulation, Planetary rotation, Ekman Layer, Sverdrup balance

2025 **University of Grenoble Alpes, Grenoble, France**, *2rd year master student*, Practical course in Turbulence, diffusion and transport - 5h
Log Layer, Turbulent Entrainment, Ekman Layer

2024 **ENS3 Engineering school, Grenoble, France**, *3rd year undergraduate student*, Practical course in thermal physics - 20h
Forced convection, Charge losses, Conduction, Heat transfer